

MyD88/TLR9 mediated immunopathology and gut microbiota dynamics in a novel murine model of intestinal graft-versus-host disease

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ABSTRACT

Background The bacterial microflora aggravates graft-versus-host-disease (GvHD) after allogeneic stem cell transplantation, but the underlying mechanisms of manifestations of intestinal GvHD (iGvHD) in the gut remain poorly understood.

Aim To analyse the gut flora composition and the impact of bacterial sensing via Toll-like receptors (TLRs) in iGvHD.

Methods By mimicking clinical low-intensity conditioning regimens used in humans, a novel irradiation independent, treosulfan and cyclophosphamide-based murine allogeneic transplantation model was established. A global survey of the intestinal microflora by cultural and molecular methods was performed, the intestinal immunopathology in TLR-deficient recipient mice with iGvHD investigated and finally, the impact of anti-TLR9 treatment on iGvHD development assessed.

Results The inflammatory responses in iGvHD were accompanied by gut flora shifts towards enterobacteria, enterococci and *Bacteroides/Prevotella* spp. Analysis of iGvHD in MyD88^{-/-}, TRIF^{-/-}, TLR2/4^{-/-}, and TLR9^{-/-} recipient mice showed that bacterial sensing via TLRs was essential for iGvHD development. Acute iGvHD was characterised by increasing numbers of apoptotic cells, proliferating cells, T cells and neutrophils within the colon. These responses were significantly reduced in MyD88^{-/-}, TLR2/4^{-/-}, TRIF^{-/-} and TLR9^{-/-} mice, as compared with wild-type controls. However, TRIF^{-/-} and TLR2/4^{-/-} mice were not protected from mortality, whereas TLR9^{-/-} mice displayed increased survival rates. The important role of TLR9-mediated immunopathology was independently confirmed by significantly reduced macroscopic disease symptoms and colonic apoptosis as well as by reduced T-cell and neutrophil numbers within the colon after treatment with a synthetic inhibitory oligonucleotide.

Conclusions These results emphasise the critical role of gut microbiota, innate immunity and TLR9 in iGvHD and highlight anti-TLR9 strategies as novel therapeutic options.

INTRODUCTION

Graft-versus-host disease (GvHD) caused by a cytokine-driven immunopathology against host and bacterial antigens is still one of the major complications of allogeneic bone marrow transplantation. By targeting mainly the colon, intestinal GvHD (iGvHD) affects more than 50% of patients with GvHD and represents a leading cause of mortality.^{1–3} Several studies have pointed

Significance of this study

What is already known about this subject?

- The commensal gut flora aggravates intestinal graft-versus-host disease (iGvHD).
- Antibiotic regimens influence the severity of iGvHD.
- TLR9 agonists promote graft rejection and graft-versus-leukaemia effects.

What are the new findings?

- Presenting a novel irradiation-independent allogeneic transplantation model in mice, which mimics modern low-intensity conditioning regimens in humans.
- Providing novel insights into intestinal bacteria–host interactions in iGvHD by studying the gut flora dynamics, the role of bacterial innate immune receptors and immunopathological responses of the host.
- iGvHD development is accompanied by gut flora shifts towards proinflammatory bacterial species and is mediated by MyD88/TLR9-dependent bacterial sensing.
- TLR9 deficiency or treatment with a synthetic inhibitory oligonucleotide (iODN) ameliorates iGvHD development.

How might it impact on clinical practice in the foreseeable future?

- Anti-TLR9 strategies such as synthetic iODN might serve as promising novel therapeutic options for patients with GvHD.

towards a critical role of the bacterial flora and donor antigen-presenting cells in initiating GvHD.^{4–5} Proinflammatory molecules produced by the recipient's gut microbiota such as lipopolysaccharide (LPS), lipoproteins, peptidoglycan or bacterial DNA are suspected to have an important role in the immunological modulation of iGvHD. Supporting this hypothesis, GvHD-related mortality was significantly reduced in germ-free mice and in conventional animals after antibiotic treatment.⁶ However, the molecular mechanisms and bacteria–host interactions contributing to the life-threatening gastrointestinal manifestations of this devastating disease remain poorly understood. Novel treosulfan-based conditioning regimens established in clinical practice as alternative to total body irradiation provide the benefit of reduced

mucositis, more efficient bone marrow depletion and less venal occlusive disease.^{7,8}

In humans, experimental analysis of the contribution of gut bacteria to iGvHD is hampered by antibiotic drugs and by dynamic changes within gut bacterial communities during intestinal inflammation. In one study, a decreased GvHD incidence was observed after antibiotic gut decontamination.⁹ Recent studies in patients with inflammatory bowel diseases (IBD) have demonstrated that intestinal inflammation alters the gut flora composition and promotes overgrowth by enterobacteria.¹⁰ Increased numbers of enterobacteria, enterococci and *Bacteroides/Prevotella* spp were found in murine colitis^{11,12} and ileitis^{13,14} and are suspected to modulate innate immune responses via Toll-like receptors (TLRs).^{14–16} Mice deficient in sensing of bacterial lipoproteins, LPS and DNA, by TLR2,^{11,14} TLR4^{11,13,14} and TLR9,¹⁷ respectively, were protected from intestinal inflammation in various murine models. Adapter proteins MyD88 and TRIF are required for TLR signalling and proper TLR3/4 activation, respectively.¹⁸ LPS-resistant mice were shown to be protected from LPS-induced GvHD mortality after allogeneic transplantation.¹⁹ Furthermore, recent studies point towards a major role of TLR9-mediated sensing of bacterial DNA in the aggravation of GvHD and graft rejection.^{20–22}

Taken together, these studies suggest that reduction of tissue damage, modulation of the bacterial flora and blocking of bacterial sensing might be effective strategies to prevent or ameliorate iGvHD. However, in iGvHD neither intestinal flora nor interactions of bacterial products with the innate immune system²³ have been analysed in low-conditioning GvHD models so far.^{19–21} Therefore, we developed a murine treosulfan-based allogeneic transplantation model for iGvHD and studied both gut flora dynamics and the role of bacterial innate immune receptors in intestinal immunopathology. We used major histocompatibility complex-mismatched MyD88^{-/-}, TRIF^{-/-}, TLR2^{-/-}, TLR4^{-/-} or TLR9^{-/-} recipient mice, lacking innate immune receptors specific for the most prominent bacterial ligands in the intestines. The role of TLR9 in iGvHD was further confirmed by amelioration of iGvHD after application of the inhibitory oligonucleotide (iODN) 2088, which inhibits TLR9 activation in vitro (see online supplementary Figure S4). Mice deficient in other TLRs, including the flagellin receptor TLR5, were not analysed here. The vast majority of gut bacteria is non-motile and corresponding ligands are not produced in relevant quantities by gut bacteria.

MATERIALS AND METHODS

Mice

All animals were maintained in the facilities of the 'Forschungseinrichtung für Experimentelle Medizin (FEM)' in Berlin, Germany, under specific pathogen-free conditions. Mice deficient in TLR2, TLR4, TLR2/4 (C57BL/10ScSn (C10) background), TLR9 or TRIF (C57BL/6 (C6) background) were provided by the Max-Planck Institut für Immunbiologie (Freiburg, Germany). TLR2^{-/-}, TLR4^{-/-} and TLR2/4^{-/-} mice were backcrossed at least seven times into the C10 background as described in detail earlier.^{11,14} TLR9^{-/-} mice were backcrossed three times to the C6 background as described.²⁴ TRIF^{-/-} breeding pairs generated in the C6 background²⁵ were kindly provided by Bruce Beutler (The Scripps Research Institute, La Jolla, USA). MyD88-deficient mice (C6 background) originated from the breeding stocks of the Max-Planck Institut für Infektionsbiologie (Berlin, Germany). Balb/c mice served as donors for transplantation experiments. Age- and sex-matched mice between 10 and 14 weeks of age were used in the experiments. All animal experiments were

approved by the appropriate state committees for animal welfare.

Conditioning, allogeneic stem cell transplantation and treatment with inhibitory oligonucleotides

Three days (day -3 to day -1) before transplantation, mice received 2000 mg treosulfan (Medac, Wesel, Germany) per kg body weight intraperitoneally (IP) daily.²⁶ The day before transplantation (day -1), 200 mg cyclophosphamide (Baxter, Deerfield, USA) per kg body weight were administered IP. On the day of transplantation (day 0), mice received 5×10⁶ bone marrow cells and 3×10⁶ spleen cells from Balb/c donors intravenously. To investigate the effect of inhibitory oligonucleotides in vivo, 200 µg of the iODN-2088 (5'-TCC TGG CCG GGA AGT-3', Alexis Biochemicals, Lörrach, Germany,²⁷) was injected IP on day +2 post-transplant.

Assessment of macroscopic symptoms of disease and chimerism analysis

Body weight and clinical symptoms of GvHD were assessed daily. The disease index was scored according to a published protocol.^{19,28} On days 10 and 28 post-transplant, and at the end of experiments, respectively, engraftment was determined by chimerism analysis (see online supplemental material). A fraction of >90% H-2D^d positive cells was defined as full chimerism, <20% H-2D^d positive cells was considered to be transplant failure. Values between 20% and 90% were defined as mixed chimerism.

Sampling procedures and histological scoring

Mice were killed by isoflurane treatment (Abbott, Wiesbaden, Germany). Biopsy specimens from stomach, jejunum, ileum and colon were taken, immediately fixed in 5% formalin and embedded in paraffin. Histological scoring was assessed on 5 µm thin sections stained with haematoxylin and eosin (H&E) and evaluated according to a standardised protocol²⁸ in triplicate by independent evaluators (AN, CL and RP).

Immunohistochemistry

In situ immunohistochemical analysis of intestinal paraffin sections was performed as described previously.¹¹ Primary antibodies against CD3 (#N1580, Dako, Glostrup, Denmark, dilution 1:10), myeloperoxidase-7 (MPO7, # A0398, Dako, 1:10000), FOXP3 (FJK-16s, eBioscience, San Diego, USA, 1:100), Ki-67 (TEC3, Dako, 1:500) and cleaved caspase-3 (Asp175, Cell Signaling, Danvers, USA, 1:200) were used.

Analysis of the intestinal microflora

Molecular detection, biochemical identification and cultural analyses of luminal bacterial communities from stomach, jejunum, ileum and colon were performed as previously described.^{11,13,14} Genetic fingerprints of the intestinal microflora were generated by denaturing-gradient-gel-electrophoresis.

Quantitative real-time PCR was performed according to previously published protocols^{29–32} on a LightCycler instrument 1.5 (Roche, Mannheim, Germany). Total bacteria, gamma-Proteobacteria/*Enterobacteriaceae*, *Lactobacillus* group, *Clostridium leptum* group, *Clostridium coccoides* group, *Bacteroides* group including *Prevotella* and *Porphyromonas*, mouse intestinal *Bacteroides*, *Enterococci* and *Bifidobacteria* were separately detected and quantified by using a SYBR Green I assay.

Real-time PCR and quantification of 16S rRNA gene sequences

DNA extractions from intestinal luminal contents were prepared as described previously.¹³ The DNA extracts and the

plasmids were quantified with Quant-iT PicoGreen reagent (Invitrogen, Paisley, UK) and were all adjusted to 1 ng DNA/ μ l.

The abundance of specific intestinal bacterial groups was measured by quantitative real-time PCR with group-specific 16S rRNA gene primers (Tib MolBiol, Berlin, Germany) listed in supplementary table 1. Each reaction mixture (20 μ l) was composed of 12.2 μ l water (Sigma-Aldrich, Steinheim, Germany), 2 μ l 10 $\times$ Buffer I and 2 μ l 25 mM MgCl₂ (Applied Biosystems, Darmstadt, Germany), 0.4 μ l PCR nucleotide mix (each dNTP 10 mM, Roche, Mannheim, Germany), 0.1 μ l bovine serum albumin (20 mg/ml, Fermentas, St. Leon-Rot, Germany), 0.1 μ l of a 1:200 dilution of SYBR Green I (10 000 $\times$ concentrate, Invitrogen, UK), 0.4 μ l of AmpliTaq DNA polymerase (2.5 IU/ μ l, Applied Biosystems, Darmstadt, Germany), 0.4 μ l of each of the specific primers (30 μ M, Tib MolBiol, Berlin, Germany) and 2 μ l DNA extract. The amplification started with an initial denaturation at 95°C for 2 min followed by 40 cycles of 95°C for 10 s, annealing at 60°C for 10 s and elongation at 72°C for 30 s. Fluorescence data were acquired at 80°C after each cycle. After amplification, melting curves for checking the targeted PCR products were obtained by heating from 60°C to 95°C at a rate of 0.05°C/s, with continuous fluorescence collection.

As reference for quantification, standard curves with 10-fold serial dilutions of plasmids (ranging from 2×10^8 to 2×10^2 copies) were generated for each run. The real-time PCR primers were first used to amplify cloned 16S rDNA of reference strains (see supplementary table 1). Each of the PCR products was cloned into pCR2.1-TOPO using the TOPO-TA cloning kit (Invitrogen, Paisley, UK) following the manufacturer's recommendations. The number of 16S rRNA gene copies/ng DNA of each sample was determined, not the actual bacterial numbers.

Statistical analysis

Mean values, medians, standard deviations and levels of significance were determined using appropriate tests as indicated. Two-sided probability (p) values ≤ 0.05 were considered significant. All experiments were repeated at least twice.

RESULTS

Induction of acute iGvHD in mice after low-intensity conditioning and allogeneic transplantation

We established a murine irradiation-independent transplantation model by using a reduced-intensity chemical conditioning regimen (figure 1A). C57BL wild-type (WT) recipient mice received treosulfan and cyclophosphamide before transplantation with 5×10^6 bone marrow cells and 3×10^6 splenocytes from Balb/c mice (figure 1A). The engraftment rate assessed by flow cytometry of peripheral blood was 92%. Mice without engraftment were excluded from further analyses. Within the first 4 days after transplantation, mice developed macroscopic signs of GvHD such as weight loss, decreased activity and ruffling fur (monitored by a standardised score, see 'Methods'), reaching maximum score values of 10.0 ± 2.0 around day 11 post-transplant (figure 2B). In contrast, Balb/c recipient mice used as syngeneic controls did not develop iGvHD as indicated by macroscopic scores of 2.0 ± 2.7 and virtually no mortality after transplantation (data not shown).

Histological analysis of sections from the stomach, ileum and colon of C57BL mice showed that macroscopic symptoms were accompanied by tissue damage along the entire gastrointestinal (GI) tract, which was most pronounced in the colon (figure 1B). Allogeneic graft recipient mice developed flattening of mucosal villi, severe crypt abscesses and complete crypt drop-outs, which were absent in syngeneic transplanted Balb/c recipient mice (figure 3).

Figure 1 The low-intensity conditioning model for intestinal graft-versus-host disease (iGvHD). (A) Treatment regimen and schedule for conditioning, transplantation and iGvHD development. C57BL recipient mice received treosulfan (Treo) on three consecutive days and cyclophosphamide (CY) on day 1 before transplantation (Tx) intraperitoneally. At day 0, mice were transplanted with 5×10^6 bone marrow (KM) cells and 3×10^6 splenocytes (SC) intravenously. (B) Quantification of mucosal tissue damage along the entire gastrointestinal tract of mice with acute iGvHD. H&E-stained tissue sections from the stomach, ileum and colon of C57BL/6 mice ($n=37$ for each organ group) with acute iGvHD at day 10 post-transplant were investigated by light microscopy, and tissue damage was quantified according to a standardised score described in the 'Methods' section. Individual histopathological scores and means from five pooled independent experiments are shown. (C) Cells infiltrating the colon mucosa during iGvHD. After transplantation of Balb/c cells on day 0, C57BL/10 mice were sacrificed every second day ($n=4$ each time point) and colon sections were investigated for T cells (CD3), regulatory T cells (FOXP3), neutrophilic granulocytes (MP07), proliferative (Ki-67) and apoptotic cells (caspase-3) by immunohistochemical staining. The development of iGvHD in the colon was characterised by accumulation of T cells (black solid line), apoptotic cells (grey solid line) and neutrophils (dotted black line). Numbers of regulatory T cells were not affected during iGvHD. Data shown are representative of three independent experiments.

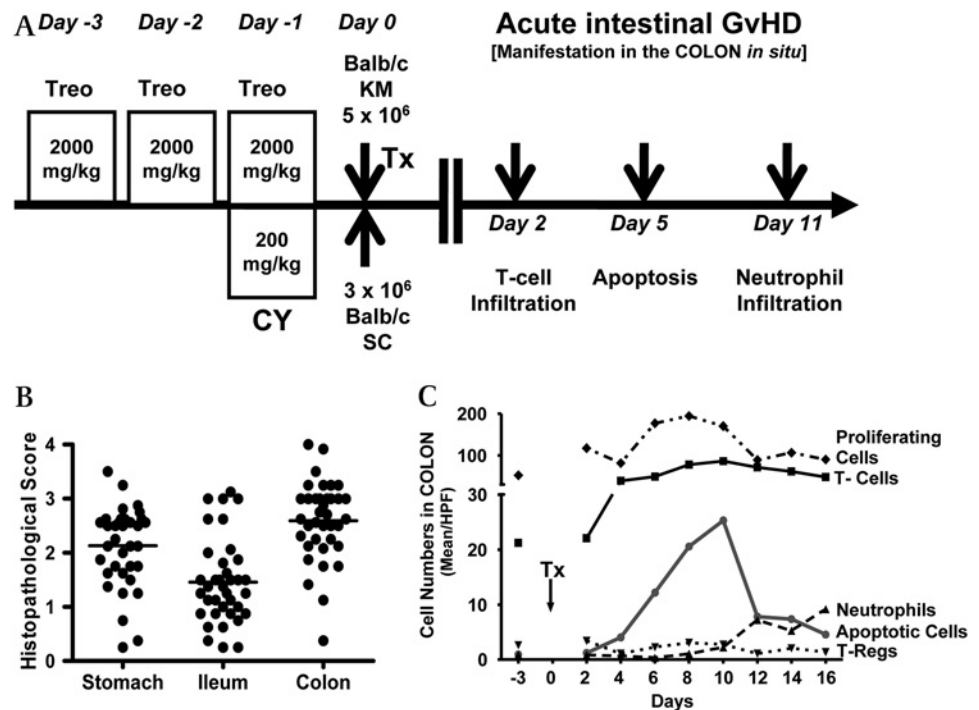
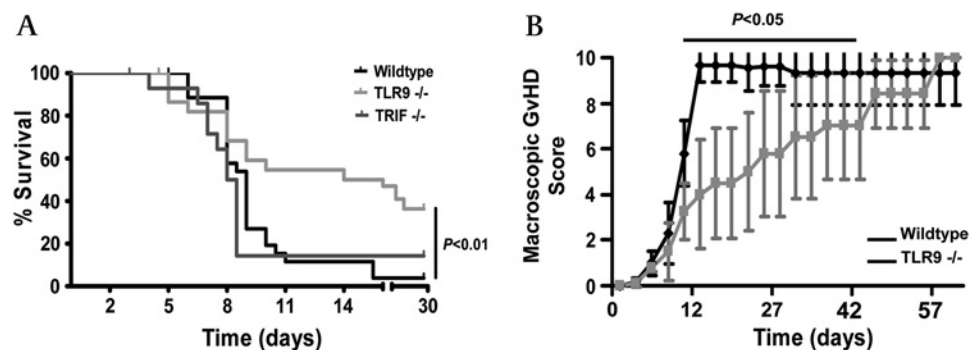


Figure 2 Bacterial sensing and survival of intestinal graft-versus-host disease (iGvHD). Survival rates (A) and macroscopic signs of iGvHD (B) were determined daily. Macroscopic signs were recorded according to a standardised scoring system (see 'Methods'). Data shown were pooled from at least three independent experiments. (A) Survival of wild-type (n=30), TRIF^{-/-} (n=24) and TLR9^{-/-} (n=27) mice was monitored until day 28 post-transplant. Significance levels were determined by Kaplan–Meier analysis and log rank test. p Value for significant differences (compared with wild-type animals) as indicated. (B) Macroscopic signs of disease in wild-type and TLR9^{-/-} mice in the course of iGvHD. Significance levels were determined by Student's t test (p values for the time span with significant differences if compared with wild-type animals are overlined by the black bar).



Specific caspase-3 staining of colon sections confirmed that numbers of apoptotic cells were significantly higher in mice with iGvHD, than in healthy control animals (figure 1C, figure 3). Kinetic analysis demonstrated that apoptotic cells could be detected in early iGvHD at day 2 post-transplant reaching maximum numbers by day 10 (figure 1C). Specific Ki-67 staining of colon sections showed that the numbers of proliferating cells were significantly elevated during iGvHD ($p<0.0001$) (figure 3).

Kinetic analysis of distinct immune cell populations in situ by staining of colon tissue sections for CD3, MPO7 and FOXP3 showed that iGvHD was accompanied by increasing numbers of T cells and neutrophils, reaching a plateau at days 9 and 12 post-transplant, respectively (figure 1C, figure 3). In contrast, populations of regulatory T cells remained constant during the onset of the inflammatory response and their numbers remained low during later stages of iGvHD (figure 3). This finding is well in line with recently observed intestinal manifestations of human GvHD where regulatory T-cell populations remained low in iGvHD.³³ These in situ results indicate that the murine reduced-intensity conditioning model mimics histopathological and immunological key features of human iGvHD within the colon.

Gut flora changes in the course of acute iGvHD

To characterise gut flora shifts during intestinal inflammation, we performed a global survey of the microflora in the colon lumen of C6-WT mice before and at day 10 after allogeneic transplantation (figure 4). Genetic fingerprints of the intestinal bacterial communities generated by PCR-denaturing-gradient-gel-electrophoresis analysis showed that the acute phase of iGvHD was characterised by a shift towards members of the *Enterobacteriaceae* family, whereas *Lactobacilli*, *Clostridia*, *Bifidobacteria* and *Bacillus* spp were less abundant (figure 4A, left), indicating that acute intestinal inflammation disrupts the gut flora and causes a loss of bacterial diversity (figure 4A, left). The strictly anaerobic communities represented mainly by *Clostridia*, *Bifidobacteria* and *Tannerella* remained unaltered during iGvHD.

These data were confirmed by quantitative real-time PCR detection of gut bacterial communities (figure 4A, right) and by cultural analysis of luminal colon contents (figure 4B). Results from quantitative RT-PCR indicated that *Bacteroides* and *Enterococcus* spp increased during iGvHD (figure 4A, right), whereas numbers of *Lactobacilli* and mouse intestinal *Bacteroides* decreased.

The total bacterial load in the cultivable bacterial populations increased significantly in the colon of diseased mice (figure 4B) as compared with healthy controls ($p<0.0001$). Numbers of commensal *E. coli*, *Bacteroides/Prevotella* spp and *Enterococci* (figure 4B) were significantly elevated during iGvHD by one to

four orders of magnitude, whereas *Lactobacilli* populations remained unaltered (figure 4B). The increase of fecal *E. coli* numbers was significantly associated with the macroscopic GvHD score and reduced long-term survival of allogeneically transplanted animals (supplementary figure S3). Cultural analyses of luminal microbial communities in more proximal parts of the GI tract showed that similar microflora shifts were seen in the small intestine (jejunum and ileum), but not in the stomach (supplementary figure S1). Taken together, cultural analyses confirmed that iGvHD alters the gut flora composition in the lower GI tract. The corresponding changes towards proinflammatory bacterial species are similar to those seen in human IBD or in murine models of intestinal inflammation.^{10–14} However, although the quantitative shifts of commensal bacteria during iGvHD were most pronounced within the *E. coli* population, peroral treatment of mice with ciprofloxacin did not ameliorate the severity of iGvHD in our reduced-intensity conditioning model (data not shown). These results suggest that iGvHD is not aggravated by a specific bacterial species, but rather by effects induced by different bacterial strains or TLR ligands.

Innate immune responses trigger apoptosis and cell proliferation during acute iGvHD

Apoptosis is commonly used as a diagnostic marker in human iGvHD. Therefore, we investigated apoptosis as well as cell proliferation during iGvHD in C6- and C10-WT mice (figure 1A) as well as in corresponding mutant strains lacking MyD88, TRIF, TLR9, TLR2, TLR4, or TLR2/4 (figure 5) by staining of colon sections with caspase-3 and Ki-67, respectively.

Quantitative analysis of caspase-3-positive cells in colon sections taken from mice with iGvHD (figure 5) displayed a significant reduction in apoptotic cells in MyD88^{-/-} and TLR9^{-/-} mice ($p<0.0001$), as compared with WT controls (figure 5A). In TLR2^{-/-}, TLR4^{-/-} (not shown) and TLR2/4^{-/-} mice, however, numbers of apoptotic cells were similar to those seen in WT animals with iGvHD (figure 5B).

Numbers of Ki-67-expressing cells were increased in the colons of TLR9^{-/-} mice, as compared with WT animals with iGvHD ($p<0.005$), whereas mice lacking MyD88, TRIF, TLR2 and TLR4 displayed proliferating cell populations similar to the respective WT controls (figure 5C, D).

Impact of bacterial sensing on the immune cell populations in the inflamed colon

Analysis of immune cell populations in the acute stage of iGvHD confirmed that MyD88-mediated sensing of gut bacteria by TLRs was involved in the initiation of the iGvHD-mediated immunopathology (figure 6). Immunohistochemical

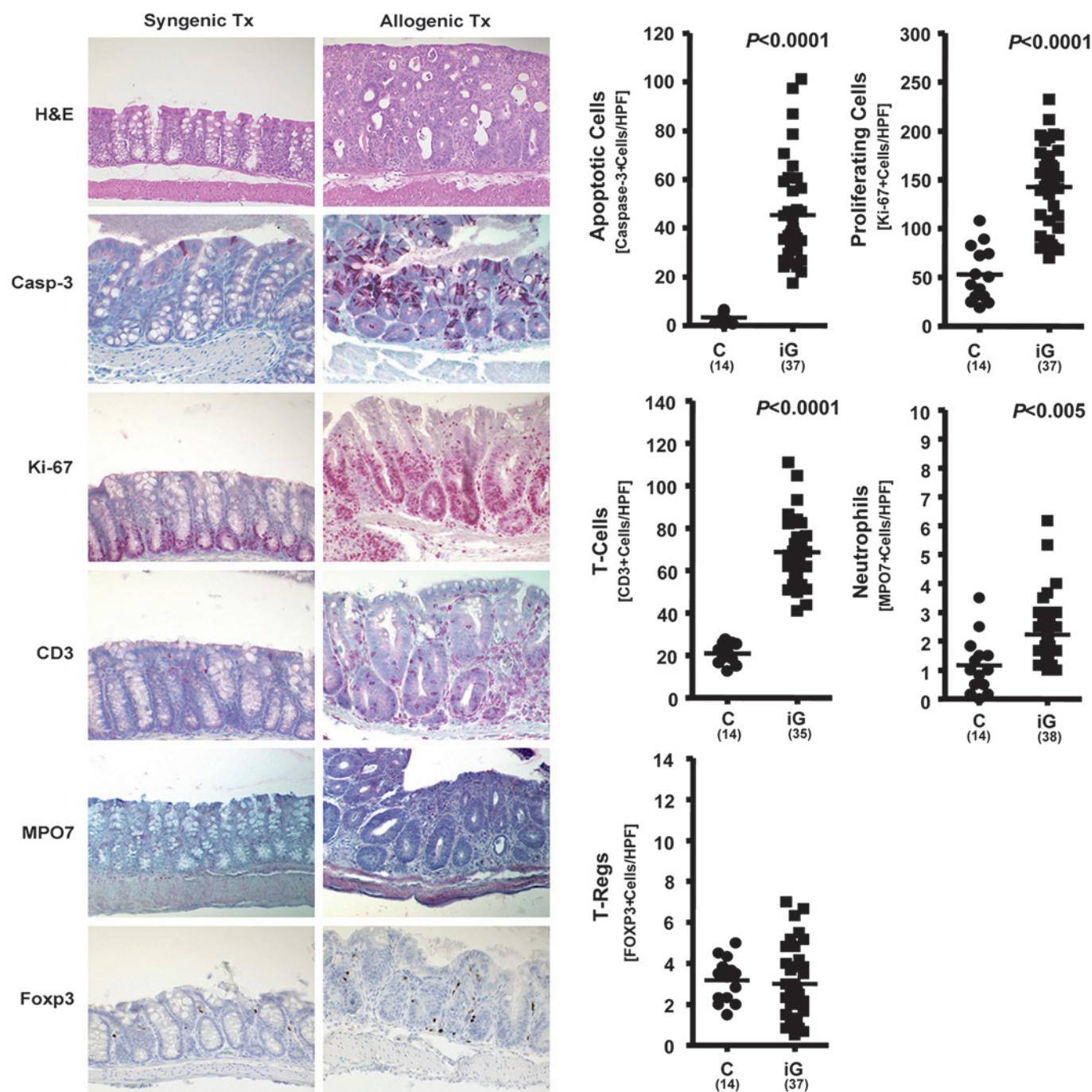


Figure 3 Defined cell populations in the colon of mice with intestinal graft-versus-host disease (iGVHD). Numbers of cells were determined in the colon of healthy control mice (C, $n=14$, filled circles) and mice with iGVHD at day 10 post-transplant (iG, $n \geq 35$, filled squares) by counting in immunohistochemically stained tissue sections from at least six representative high-power fields (HPF, $\times 400$ magnification) per animal. Representative stained tissue sections of control animals and mice with iGVHD are shown in the left panel. Apoptotic and proliferating cells were detected by staining for caspase-3 or Ki-67, respectively. T cells, neutrophilic granulocytes, and regulatory T cells (T-Regs), were detected by CD3, myeloperoxidase-7 (MPO7), or FOXP3 staining, respectively. Medians and significance levels (p values, as compared with healthy controls) determined by Mann-Whitney U test are indicated. Data shown were pooled from four independent experiments.

staining showed that CD3+ T-cell numbers were reduced in the colon of MyD88^{-/-} ($p < 0.0001$), but not TRIF^{-/-} mice at day 10 post-transplant (figure 6A). This difference could be attributed to TLR2, 4 and 9 signalling, as indicated by significantly reduced numbers of T cells in TLR2^{-/-}, TLR4^{-/-}, and TLR2/4^{-/-} (figure 6B), as well as TLR9^{-/-} mice (figure 6A), as compared with the corresponding WT animals. Numbers of MPO7+ neutrophilic granulocytes were significantly lower in

the colon of MyD88^{-/-}, TRIF^{-/-} and TLR9^{-/-} animals (figure 6C), but not in TLR2^{-/-}, TLR4^{-/-} and TLR2/4^{-/-} animals (figure 6D), as compared with the respective WT controls with acute iGVHD.

Reduced mortality and macroscopic signs of disease in mice lacking TLR9

To evaluate the relevance of bacterial sensing for the outcome of iGVHD, we investigated the potential impact of these receptors

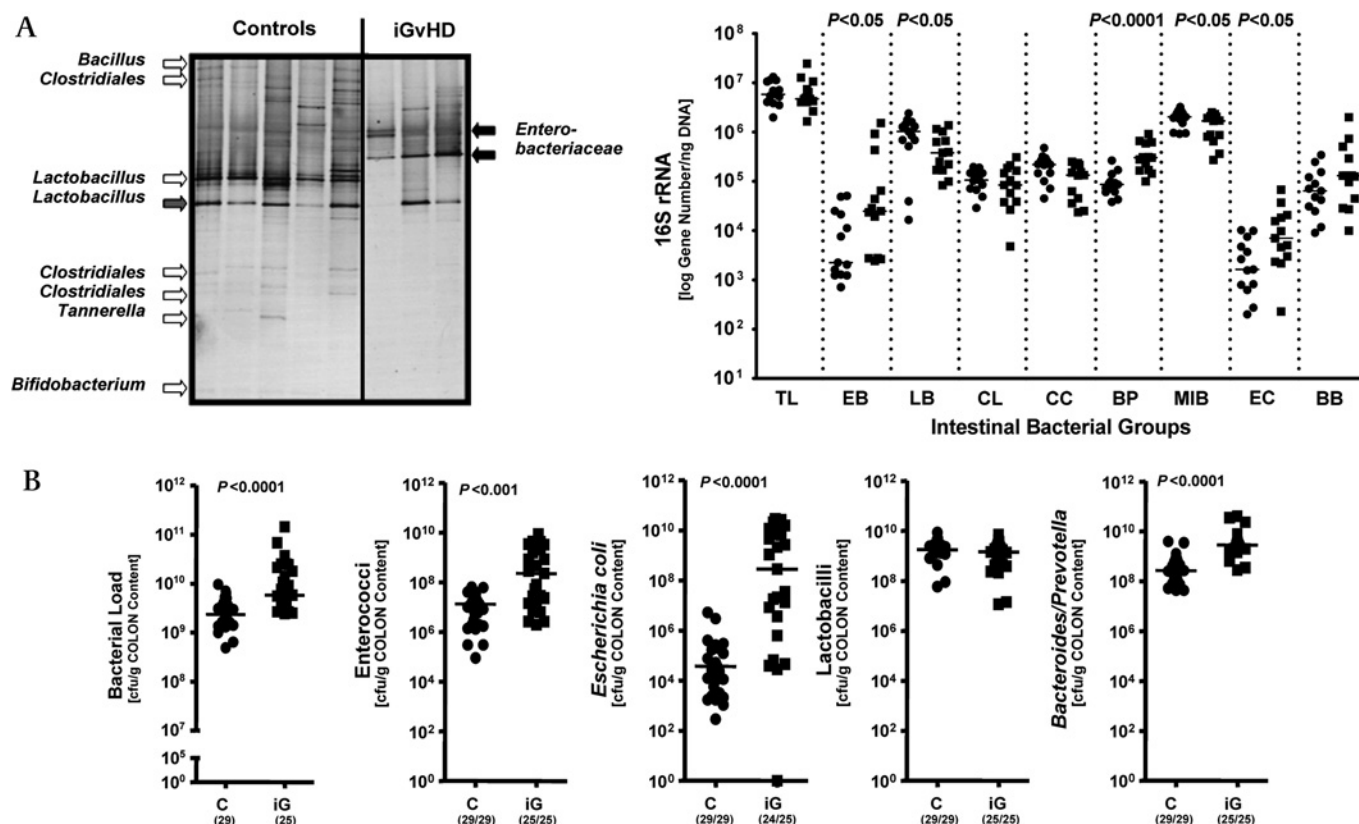


Figure 4 Composition of the intestinal bacterial microflora in mice with intestinal graft-versus-host disease (iGvHD). Detailed analysis of the microbial communities in healthy animals and mice with iGvHD was performed by denaturing-gradient-gel-electrophoresis (DGGE) and real-time-PCR (A) or by culture (B) on appropriate selective media. (A) Molecular fingerprints (left) and quantitative analysis by real-time-PCR (right) of bacterial groups in the colon flora of healthy control animals and mice with iGvHD (day 10 post-transplant). Molecular profiles were generated by PCR-based DGGE analysis of total DNA isolated from luminal colon contents. Bacterial species were identified by comparative sequence analysis of 16S rRNA gene fragments amplified from DNA eluted from respective DNA bands. The DGGE profiles are representative for at least three mice per group and experiment. Real-time PCR analysis of gut bacterial groups in luminal colon contents of healthy mice (filled circles) and mice with iGvHD (filled squares) was performed as described in the 'Methods' section. Total bacteria (TL), gamma-Proteobacteria/Enterobacteriaceae (EB), *Lactobacillus* group (LB), *Clostridium leptum* group (CL), *Clostridium coccoides* group (CC), *Bacteroides* group including *Prevotella* and *Porphyromonas* (BP), mouse intestinal *Bacteroides* (MIB), *Enterococci* (EC) and *Bifidobacteria* (BB) were separately detected and quantified by real-time PCR. Medians and significance levels (p values, as compared with healthy animals) determined by Mann–Whitney U test are indicated. Data shown are pooled from three independent experiments. (B) Total bacterial counts and numbers of *Enterococci*, *E. coli*, *Lactobacilli* and *Bacteroides/Prevotella* spp in luminal colon contents from healthy control animals (C, n=29) and from mice with iGvHD (iG, n=25) at day 10 post-transplant were determined by detection of colony forming units (cfu) on appropriate culture media (see 'Methods'). Bacterial species were identified by biochemical analysis and reconfirmed by comparative sequence analyses of 16S rRNA genes. Medians and significance levels (p values, as compared with control animals) determined by Mann–Whitney U test are indicated. Data shown were pooled from four independent experiments.

on survival. Balb/c stem cells were grafted into either C10-WT, TLR2^{-/-}, TLR4^{-/-}, TLR2/4^{-/-}, or C6-WT, TRIF^{-/-}, TLR9^{-/-} mice and survival rates determined (figure 2 and supplementary figure S2). Results from chimerism analyses showed no significant differences between the engraftment rates of the respective mice strains (data not shown).

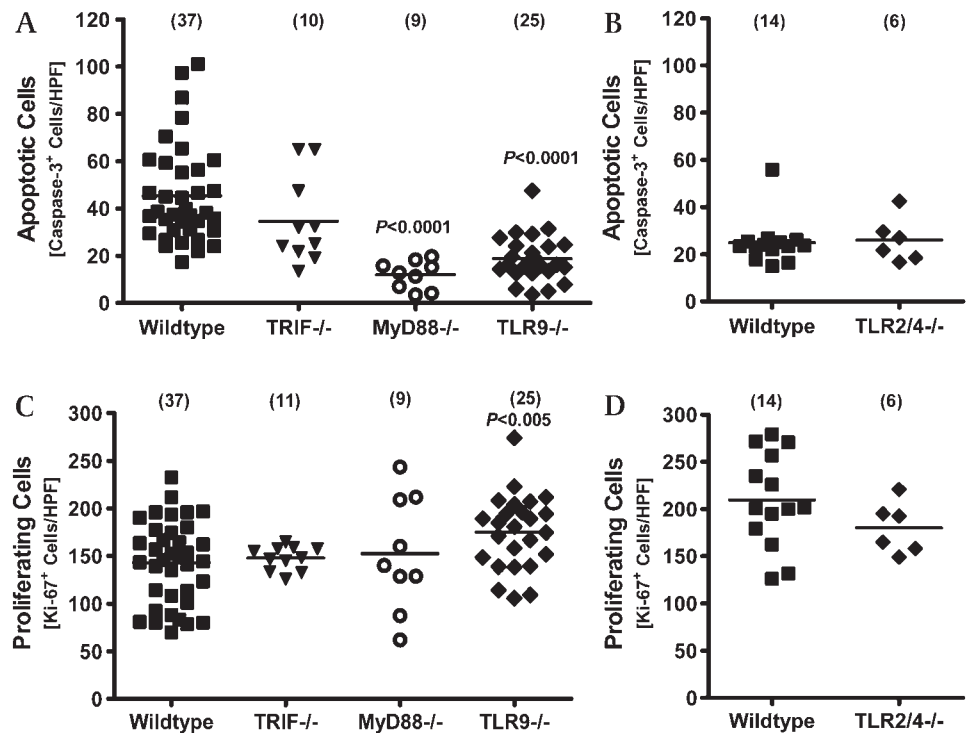
After induction of iGvHD, TLR2^{-/-}, TLR4^{-/-} and TLR2/4^{-/-} mice displayed survival rates similar to WT animals (supplementary figure S2). In contrast, TLR9^{-/-} mice showed significantly reduced iGvHD mortality (p<0.01) as compared with diseased WT controls (figure 2A), which was further confirmed by less pronounced macroscopic signs of disease in TLR9^{-/-} mice over time (figure 2B).

Amelioration of TLR9-mediated apoptosis in iGvHD after treatment with inhibitory oligonucleotide

While recent investigations confirmed that CpG oligonucleotides (ODNs) markedly accelerated GvHD mortality via TLR9 activation,²¹ a beneficial role of inhibitory ODNs (iODNs) in

iGvHD has not been described so far. To independently confirm the important role of TLR9 signalling in aggravation of iGvHD, we investigated possible iODN functions in amelioration of iGvHD. Therefore, allogeneically transplanted recipient mice were treated with iODN-2088, which was shown to inhibit TLR9 activation of macrophages in vitro (supplementary figure S4). Recipient mice received a single injection of iODN-2088 IP at day 2 post-transplant. Because apoptosis is the most sensitive measure for iGvHD manifestations in humans, the severity of iGvHD was assessed by caspase-3 staining and counting of apoptotic cells in colon sections at day 10 post-transplant (figure 7). The degree of immunopathology was assessed by MPO7 and CD3 staining of neutrophils and T cells, respectively. The results demonstrate that iODN-2088 application ameliorated apoptosis and immunopathology caused by iGvHD, as indicated by significantly reduced numbers of apoptotic cells (p<0.005), neutrophils (p<0.005) and T cells (p<0.05) in colon sections of iODN-treated animals as compared with placebo-treated controls with severe iGvHD (figure 7).

Figure 5 Influence of bacterial sensing by host receptors on apoptosis and cell proliferation in the colon in situ. The average number of apoptotic cells (positive for caspase-3, A and B) or proliferating cells (positive for Ki-67, C and D) from at least six high-power fields (HPF, $\times 400$ magnification) per animal were determined microscopically in immunohistochemically stained colon sections of mice with acute intestinal graft-versus-host disease (iGvHD) (G) at day 10 post-transplant. Medians and levels of significance (p values) as compared with wild-type animals determined by Mann–Whitney U test are indicated. Data shown were pooled from two independent experiments. (A) Apoptotic cells in the colon of wild-type (n=37), TRIF^{-/-} (n=10), MyD88^{-/-} (n=9) and TLR9^{-/-} (n=25) mice with iGvHD. (B) Apoptotic cells in the colon of wild-type (n=14) and TLR2/4^{-/-} (n=6) mice with iGvHD. (C) Proliferating cells in the colon of wild-type (n=37), TRIF^{-/-} (n=11), MyD88^{-/-} (n=9) and TLR9^{-/-} (n=25) mice with iGvHD. (D) Proliferating cells in the colon of wild-type (n=14) and TLR2/4^{-/-} (n=6) mice with iGvHD.



Additional data concerning amelioration of disease symptoms by iODN-2088 treatment are summarised in supplementary figure S6 (macroscopic disease index and histopathological scores).

DISCUSSION

The low-conditioning murine model developed here mimics commonly used clinical conditioning regimens^{4,5} and is suited to unravel interplays between intestinal flora and the innate immune system in iGvHD. The rates of engraftment (>90%)

and mortality (<1%) within the first 4 days post-transplant confirmed the efficacy and tolerability of this regimen. Transplanted WT mice displayed major clinical characteristics of iGvHD manifestations in humans.³⁴ Tissue damage was most pronounced in the lower intestinal tract and characterised by cell apoptosis, flattened villi or complete crypt drop-outs, resembling key features of iGvHD histopathology in human iGvHD.³⁵ Similar to acute human iGvHD^{2–4,35,36} inflammation in murine iGvHD was characterised by increasing numbers of proliferating cells, T cells and neutrophils in the colon. This suggests that

Figure 6 Influence of host bacterial sensing on immune cell populations in the colon in situ. The average numbers of T cells (positive for CD3, A and B) and neutrophilic granulocytes (positive for myeloperoxidase-7 (MPO7), C and D) from at least six high-power fields (HPF, $\times 400$ magnification) per animal were determined microscopically in immunohistochemically stained colon sections of mice with acute intestinal graft-versus-host disease (iGvHD) at day 10 post-transplant. Medians and levels of significance (p values) as compared with wild-type animals (determined by Mann–Whitney U test) are indicated. Data shown were pooled from two independent experiments. (A) T cells in the colon of wild-type (n=35), TRIF^{-/-} (n=11), MyD88^{-/-} (n=9) and TLR9^{-/-} mice (n=25) with iGvHD. (B) T cells in the colon of wild-type (n=14), TLR2^{-/-} (n=10), TLR4^{-/-} (n=5) and TLR2/4^{-/-} (n=6) mice with iGvHD. (C) Neutrophilic granulocytes in the colon of wild-type (n=38), TRIF^{-/-} (n=11), MyD88^{-/-} (n=9) and TLR9^{-/-} (n=25) mice with iGvHD. (D) Neutrophilic granulocytes in the colon of wild-type (n=14), TLR2^{-/-} (n=10), TLR4^{-/-} (n=5) and TLR2/4^{-/-} (n=6) mice with iGvHD.

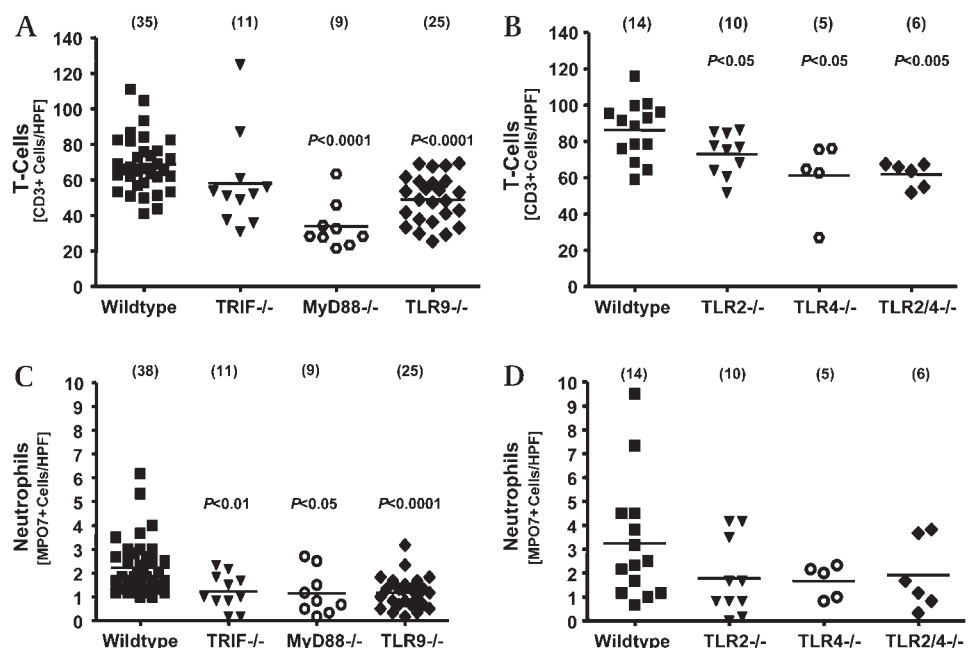
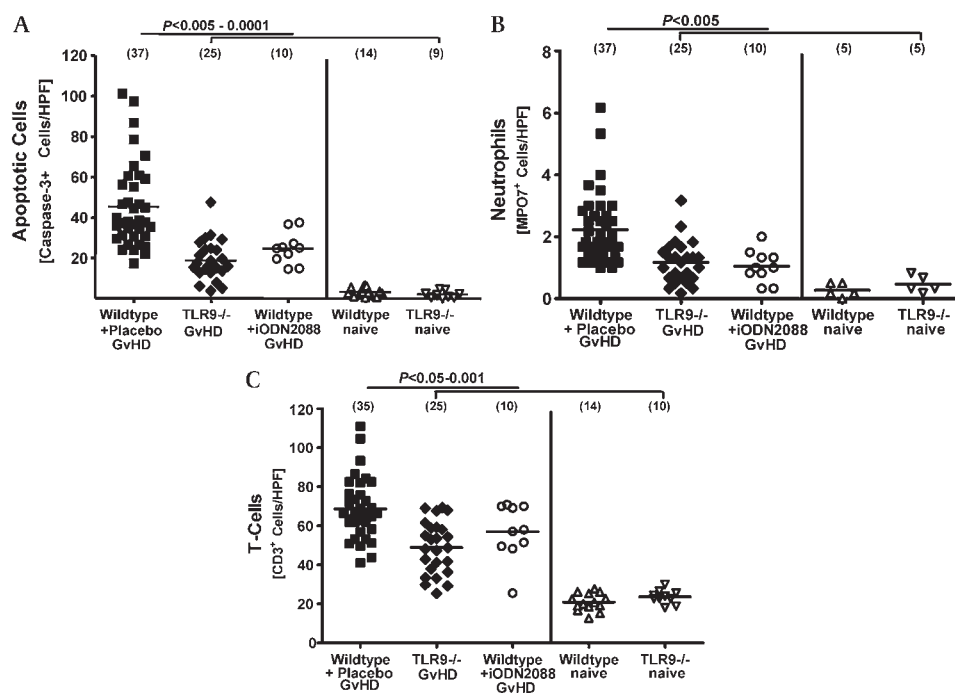


Figure 7 Influence of inhibitory oligonucleotide (iODN-2088) and TLR9 deficiency on apoptosis and immunopathology in the colon in situ. Apoptotic cells (A), neutrophils (B) and T cells (C) staining positive for caspase-3, myeloperoxidase-7 (MPO7), or CD3, respectively, in the colon of wild-type mice treated with placebo or iODN-2088 as indicated on the x-axis were compared with TLR9^{-/-} mice with intestinal graft-versus-host disease (iGvHD; day 10 post-transplant) and with naive, healthy controls (numbers of animals are given in parentheses). The average number of stained cells from at least six high-power fields (HPF, $\times 400$ magnification) per animal was determined microscopically in immunohistochemically stained colon sections. Medians and levels of significance (p values) as compared with placebo-treated, wild-type animals with iGvHD (determined by Mann–Whitney U test) are indicated. Data shown were pooled from three independent experiments.



monitoring of cell proliferation could add to the histological diagnosis of iGvHD. Regulatory T cells did not increase in the colon of mice in the course of iGvHD. This is well in line with results from human iGvHD where low numbers of regulatory T cells in colon biopsy specimens were proposed as a diagnostic marker to discriminate human iGvHD from infection.³³

Murine iGvHD was accompanied by flora shifts towards *E. coli*, *Enterococci*, and *Bacteroides/Prevotella* spp. Similar changes were seen in experimental models of intestinal inflammation^{11,13,14} and in patients with IBD,^{12,23} indicating that these alterations are caused by inflammation. This was further confirmed by the fact that gut flora shifts started with inflammation around day 7 post-transplant. Thus, bacterial flora shifts were not caused by mucositis or the conditioning regimen. The extensive overgrowth by *E. coli* indicates that fast-growing bacteria take advantage of increased sugar and oxygen concentrations induced by inflammation and tissue damage.¹¹ Notably, increased intestinal *E. coli* loads were significantly associated with GvHD mortality and intestinal pathology (supplementary figure S3). This is in line with the proinflammatory potential of *E. coli*, enterococci and *Bacteroides* spp in various models of intestinal inflammation.^{11,13,37}

Our results demonstrate that MyD88-dependent TLR9 signalling of bacterial DNA is essential for induction of apoptosis and immune cell infiltration in the colon during iGvHD. The proinflammatory potential of TLR9 activation in the allogeneic setting was confirmed by in vitro results, which showed that CpG-DNA induces T-cell proliferation in allogeneic murine mixed-lymphocyte reactions (supplementary figure S5). Other TLR ligands were shown to stimulate T-cell-mediated autoimmunity by induction of T-cell proliferation in response to bacterial LPS in vitro,³⁸ and the survival of purified CD4⁺ T cells was prolonged by administration of TLR2 and TLR4 ligands, as shown earlier by other investigators.³⁹ This is well in line with the reduction of T-cell numbers in the colon of the MyD88^{-/-}, TLR2^{-/-}, TLR4^{-/-} and TLR2/4^{-/-} mice with iGvHD. The role of TLR9 in the onset of iGvHD is in agreement with the observations that human GvHD can be ameliorated by antibiotic gut decontamination,⁹ and the TLR9-mediated activation of B-cell populations

in patients with iGvHD.⁴⁰ We used the beneficial effects of iODNs in experimental colitis¹⁷ as a basis to independently confirm the role of TLR9 in iGvHD. Results indicate that iODN-2088 treatment reduces apoptosis, the most sensitive measure for iGvHD in the colon in situ. This needs to be further investigated because therapeutic effects of iODNs in experimental models of intestinal inflammation are complex and in part conflicting.¹⁷

In conclusion, the novel murine model mimics many important aspects of human iGvHD and thus is useful for a more detailed understanding of gut bacteria–host interactions in iGvHD. Together, with the observation that iODN-2088 ameliorated iGvHD in this model, the findings may lead to alternative therapeutic options.

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